

Deep Learning Methods for Generating Tiger Videos

Your Name

1 Data

To generate realistic tiger videos, the dataset consists of video clips of tigers collected from wildlife footage and open-source datasets such as ImageNet-Vid, YouTube-VIS, or other animal video datasets. The video clips are preprocessed by:

- Resizing all frames to 128×128 pixels,
- Extracting fixed-length clips (16 frames per clip),
- Normalizing frame values and performing data augmentation such as random horizontal flipping, color jitter, and temporal jittering.

No labels are required since the generation task is unconditional. The model learns the underlying distribution of tiger videos in a fully unsupervised manner.

2 Network Architecture

2.1 Baseline: Video GAN (MoCoGAN)

The baseline model uses a GAN-based approach, where the generator maps a random noise vector into a sequence of video frames, and the discriminator evaluates the realism of both individual frames and temporal consistency across frames.

Generator:

- Takes a content latent vector z_c (controls appearance) and a motion latent vector sequence $z_m(t)$ (controls temporal dynamics),
- Uses 3D convolutions or ConvLSTM layers to generate a video clip.

Discriminator:

- 2D CNN for per-frame realism,
- 3D CNN for temporal consistency.

2.2 Main Model: Latent Video Diffusion

The main model employs a diffusion-based approach to improve temporal consistency and visual quality.

Architecture:

- **VAE Encoder:** Encodes each frame into a latent space,
- **Temporal U-Net / 3D U-Net with Attention:** Performs diffusion denoising in latent space across frames,
- **Decoder:** Reconstructs video frames from latent representations.

The model is trained to iteratively remove noise added to latent representations of video frames, learning the underlying distribution of tiger video clips.

3 Other Resources

- **Hardware:** 2–4 GPUs (e.g., NVIDIA RTX 4090 or A100), with at least 24GB memory per GPU.
- **Software:** PyTorch, HuggingFace Diffusers, PyTorch Lightning.
- **Pretrained models:** Pretrained VAE or image encoder backbones can accelerate diffusion model training.
- **Evaluation metrics:** Fréchet Video Distance (FVD), Frame-level FID, LPIPS for temporal perceptual similarity.

4 Design Thoughts

Starting from a GAN-based baseline ensures rapid prototyping and feasibility, while the latent video diffusion model provides high-quality generation with improved temporal consistency. The unconditional setting relies on tiger video data to implicitly guide the generation. Future extensions may include class-conditioned or text-guided generation for more controllable outputs.

GAN models may suffer from flickering and instability, whereas diffusion models are more stable but computationally intensive. Comparing these two approaches allows assessment of trade-offs between quality, temporal consistency, and training efficiency.